

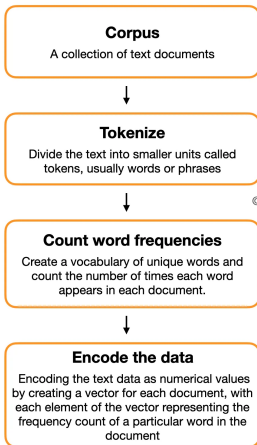
PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

Christopher Zorn

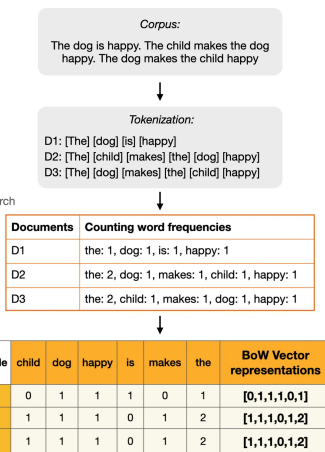
September 30, 2026

Origins: “Bag of Words” Models

Bag of Words Model explanation



Example



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Classic example:

$$\text{"Queen"} = \text{"King"} - \text{"Man"} + \text{"Woman"}$$

An even better example (via [Michael Burnham](#)):

	Animal	Fluffy	Pet	Dangerous	Food
Cat	.98	.94	.87	.30	.15
Dog	.98	.87	.95	.35	.15
Pig	.95	.20	.45	.26	.60
Carrot	.01	.02	.01	.01	.97
Rock	.01	.02	.43	.25	.01

Bag of words: a document is a pile of word counts

- Order is ignored: “dog bites man” = “man bites dog”

Embeddings: each word is a point in “meaning space”

- Similar words sit near each other (*king* / *queen*)
- Learned from data (Raschka: the computer “figures out” the pattern)

But: every word still gets *one* point, no matter where it appears

The Problem With Cats

Allamar (2021) example:

- “She deposited her paycheck at the **bank**.” vs. “They had a picnic on the river **bank**.”
- “Pick up that **stick**” vs. “**Stick** with it” vs. “That joke didn’t **stick**.”

Or consider:

“This is my cat Miles.”

“Who let the cat out of the bag?”

“Dr. Gonzalez is one cool cat.”

“That CAT dug the trench in no time.”

“It’s raining cats and dogs.”

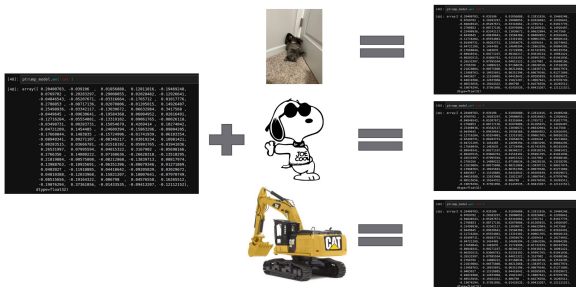
A single, fixed embedding averages across all of these.

The Problem: Words Change Meaning

Humans resolve this instantly by looking at the *surrounding words*

- Allamar: newer models look at “the entire sentence before assigning each word... an embedding”

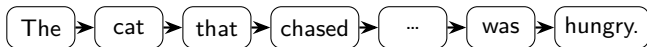
Global embeddings → Contextual embeddings...



(source)

The Old Way: One Word at a Time

Pre-2017, the best language models (“recurrent” networks) read like we do: left to right, one word at a time



→ Each step carries a running “memory” of what came before

Problems:

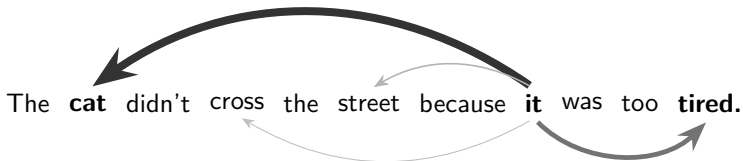
1. **Forgetting:** by the time it reaches *hungry*, the model has trouble “remembering” the *cat*
2. **Slow:** step 7 can’t start until step 6 is done; training on very large amounts of text takes too long

2017: “Attention Is All You Need”

A team at Google proposed a new design: the **transformer**

- Core idea: **look at all the words at once**, and let each word decide which other words matter to it
- Lohani’s metaphor: attention is “a spotlight” that focuses on the most relevant parts of a sentence
- Two payoffs:
 - **Better context:** distant words are just as easy to connect as neighboring ones
 - **Speed:** all words processed in parallel $\Rightarrow$ can train on vastly more text

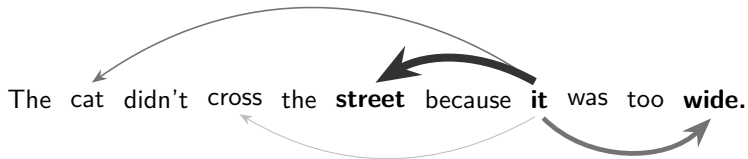
Which words should “it” pay attention to?



- Thicker arrow = higher weight = more attention
- Here, “it” borrows meaning mostly from “cat”

Attention, Pictured

Which words should “it” pay attention to?



- Change “tired” to “wide” and the spotlight shifts to “street”
- In transformers, *every word does this for every other word, all at once*
- The “weights” are *learned from data*

Many Spotlights at Once

Transformers run many attention “heads” side by side

- Each can pick up a different kind of relationship, e.g.:
 - who is doing what to whom
 - emotional tone
 - timeline of events
- These layers are *stacked*: later layers build on earlier ones

Note: these labels are human interpretations. Researchers often *can't* say cleanly what a given head is doing.

A cool visual explainer is here:

<https://poloclub.github.io/transformer-explainer/>.

Where Does the Knowledge Come From?

Step 1: Pre-training on enormous amounts of *unlabeled* text

- Fill in the blank (BERT): “The _____ barked at the cat.”
- Guess the next word (GPT): “The dog barked at the _____”
- No human labeling needed; the text supplies its own answers (Raschka: **self-supervised** learning)

Step 2: Fine-tuning for a specific job via *transfer learning*

- Spam detection, sentiment, question answering, chatting...
- Anyone can download a pre-trained model and adapt it:
Alammar calls this NLP’s “ImageNet moment”

Why it matters: one expensive general model → many cheap specialized uses

Putting It Together

Approach	What a word “is”	Context?
Bag of words	a count	none
Embeddings	a fixed point	blurred across uses
Transformers	changes by sentence	yes, via <i>attention</i>

- Parallelism $\Rightarrow$ scale $\Rightarrow$ GPT, BERT, Claude, Gemini...
- Same design now used for images, audio, video, even DNA
- **Next:** Used for recommendations, rankings, conversations
- **Later:** what happens when a transformer is trained to *generate* words/sounds/images?